

Doctoral Research Statement of Intent

Topologically Grounded Diffusion and Causal Spatial Representations for Bidirectional Sign Language AI

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Proposed Primary Supervisor: Dr. Azade Farshad · Recruitment Call: Autumn 2026 ELLIS Doctoral Fellowship

Live 15-DOF Prototype: [WebGL Demo](#) | Full 14-Page Proposal: [Direct PDF](#) | Code: [GitHub Repo](#)

EXECUTIVE RESEARCH VISION & CORE HYPOTHESIS AT A GLANCE:

- ▶ **Societal Impetus:** Over 70M Deaf signers face chronic communication barriers. Under the WHO standard of health as physical, mental, and social well-being, automated continuous sign language AI serves as a vital assistive digital twin.
- ▶ **Core Technical Thesis:** End-to-end unconstrained generative models fail due to *spatial drift* and *minimal-pair smearing*. We resolve this via a **Neuro-Symbolic Foundation Paradigm**: closed-form 15-DOF procedural scaffolding bounds continuous residual diffusion via Riemannian manifold constraints (extending Dr. Farshad's *SCOPE*), coupled with an autoregressive **Spatial-SignLLM** tracking 3D coordinate loci in an external Causal Spatial Memory Bank (M_t).
- ▶ **Execution & Compute:** 4-year cumulative PhD (3 flagship papers: **CVPR** → **ACL** → **NeurIPS**) scaled on CSC's **EuroHPC LUMI supercomputer** (AMD MI250X nodes) and co-designed with the Finnish Association of the Deaf (*Kuurojen Liitto ry*).

1 Problem Formulation & Societal Accessibility Impetus

Over **70 million Deaf individuals** worldwide communicate primarily through natural sign languages [4, 13]. The systemic absence of continuous, qualified interpretation in civic life, healthcare consultations, and everyday commerce severely compromises communicative autonomy and creates chronic social isolation. Under the World Health Organization (WHO) definition of health, linguistic exclusion represents a profound social determinant of well-being. In Finland and the European Union, equal communicative access is legally mandated by Finland's Sign Language Act (*Viittomakielilaki 352/2015*) and the European Accessibility Act (Directive 2019/882).

Sign language is not a 1D sequence of acoustic phonemes; it is an asynchronous, multi-channel **4D spatio-temporal broadcast** in 3D Euclidean space over time [4]: (1) 15-DOF finger articulations and palm orientations; (2) non-manual facial signals (eyebrows AU1/AU2/AU4) and gaze vectors; and (3) a dynamic system of **Referential Loci (R-loci)**—virtual 3D spatial anchors projected into the signing volume to track discourse referents and govern directional agreement verbs [8, 10].

Current machine learning paradigms exhibit critical failure modes in this domain:

- ▶ **Forward Production (Text-to-Sign, SLP):** End-to-end discrete diffusion models (e.g., Pose-VQVAE [15]) suffer from codebook quantization blur, smearing centimeter-scale minimal pairs (e.g., confusing MOTHER at the chin with FATHER at the forehead). Conversely, rule-based procedural avatars guarantee phonological compliance but exhibit unnatural rigidity and lack biological co-articulation.
- ▶ **Backward Translation (Sign-to-Text, SLT):** Under streaming constraints (< 300 ms), sliding temporal context windows truncate historical video, purging persistent spatial loci. This induces *Spatial State Collapse* [10], leaving models unable to resolve directional agreement verbs. Furthermore, skeleton-only translators lack environmental pixels to disambiguate polysemic depicting classifiers (e.g., resolving whether handshape CL:C denotes a cup, pipe, or container) [14].

This doctoral research resolves these bottlenecks through an **Autoregressive Spatial Sign Language Large Model (Spatial-SignLLM)** foundation architecture coupled with deterministic procedural scaffolding and topologically constrained diffusion.

2 Proposed Technical Methodology

2.1 1. Forward Pass: Procedural Scaffolding with Continuous Residual Diffusion

Rather than forcing a neural denoiser to hallucinate high-dimensional 3D kinematics from scratch under global data scarcity (~ 80 hours), we decouple symbolic syntax from physical naturalization across three stages:

1. **Symbolic Syntax & Spatial Allocation:** Spoken text is parsed into sign Time-Topic-Comment (TTC) order. Semantic entities are assigned 3D coordinate anchors (Locus_1 at the sternum, $\text{Locus}_{3a}/\text{Locus}_{3b}$ laterally).
2. **Deterministic Procedural Scaffolding:** Formulated with academic mentorship from Dr. Marco Marini (Sapienza University of Rome) and empirical lexical corpora (ASL-LEX 2.0 [2, 12], 3D-LEX v1.0 [11]), signs are assembled from atomic sub-lexical primitives. A deterministic 52-joint SMPL-X scaffold $\hat{\mathbf{P}} \in \mathbb{R}^{T \times 52 \times 3}$ is generated via 3-phalanx digit kinematic chains and analytical 2-bone arm Inverse Kinematics (IK) with circular swivel-plane projection and a singularity escape regularizer ($\|\hat{v}_{\text{swivel}} \times \hat{u}\| + \epsilon$).
3. **Continuous Residual Diffusion:** The scaffold provides a continuous geometric prior ($\mathbf{X}_{\text{final}} = \hat{\mathbf{P}} + \Delta\mathbf{X}$). A Temporal Diffusion Transformer (DiT) denoises residual micro-displacements $\Delta\mathbf{X}$, governed by our geometry-preserving objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diffusion}} + \lambda_{\text{geom}}(\mathcal{L}_{\text{landmark}} + \mathcal{L}_{\text{manifold}}) \quad (1)$$

where $\mathcal{L}_{\text{landmark}}$ enforces localized spatial tolerance bubbles ($r_{\text{tol}} \leq 5$ mm) to protect minimal pairs; and $\mathcal{L}_{\text{manifold}}$ enforces bone-length graph invariance and anatomical limits (extending *SCOPE* [6]), suspended during contact ($\gamma_{\text{contact}}(t)$) for crisp emphatic taps.

2.2 2. Backward Pass: Dual-Rate Perception and Causal Spatial Memory (M_t)

1. **Dual-Rate Stream & Active Visual Saccades:** We decouple continuous processing into a 60 Hz kinematic stream (15-DOF finger angles) and a 1 Hz vision encoder extracting scene context ($\mathbf{z}_{\text{scene}}$) from monocular RGB video (with active depth / LiDAR suggested as an edge-client extension). When decoding entropy spikes, an active visual saccade fetches a localized crop ($\mathbf{v}_{\text{patch}}$), resolving classifier ambiguity (e.g., CL:C) with $> 95\%$ compute savings.
2. **Causal Spatial Memory Bank (M_t):** An external stateful register $M_t = \{\langle \mathbf{x}_k, \mathbf{e}_k, \alpha_k, \tau_k \rangle\}_{k=1}^K$ tracks active 3D referents in Euclidean space. Bound via convergent gaze and manual pointing vectors, active loci decay exponentially. When directional agreement verbs occur, spatial θ -role attention over M_t resolves Source and Goal loci for path verbs, while a normalized aiming vector $\vec{v}_{\text{aim}}$ resolves non-displacement orientation verbs (LOOK-AT, WARN). A verb-class mapping $\mathcal{M}(v)$ [9] maps them to syntactic arguments, resolving regular, backward (INVITE), and orientation agreement immune to temporal window truncation.

2.3 3. The Spatial-SignLLM: 3D Spatial-Lexical Foundation Modeling

Drawing on embodied AI foundations (e.g., Google RT-2 [1], PaLM-E, SpatialVLM), bidirectional coupling is established via an autoregressive **Spatial Sign Language Large Model (Spatial-SignLLM)**. Crucially, rather than generating continuous 60 Hz joint torques (which induces token explosion), the LLM functions as a **1–2 Hz macro-instruction compiler**: it emits structured phonological primitives and composes depicting classifiers for out-of-vocabulary terms (e.g., cakepop) grounded in perceptual loci ($\mathcal{V}_{\text{total}} = \mathcal{V}_{\text{text}} \cup \mathcal{V}_{\text{phono}} \cup \mathcal{V}_{\text{spatial}}$), parameterizing the downstream 60 Hz procedural scaffold and residual diffusion denoiser. In the backward pass, online perceptual tokens and memory anchors from M_t are decoded into fluent text while tracking discourse agreement without state collapse.

3 Methodological Synergy with Dr. Azade Farshad's Group & ELLIS Finland

While Dr. Farshad's group has established international leadership in medical image analysis and surgical data science, my research adapts this geometric and generative toolkit to an urgent, complementary domain: **continuous sign language AI as an assistive communication digital twin**. This initiative directly aligns with ELLIS Institute Finland's core strategic priority, **AI for Health & Society**, addressing communication equity as a vital determinant of mental health and social inclusion.

Rather than viewing sign language through a generic NLP lens, this proposal builds directly upon two foundational pillars of Dr. Farshad's research:

1. **Scene-Graph Structured Generative Conditioning (Extending SceneGenie):** In *SceneGenie* [5], Dr. Farshad demonstrated that complex visual generation is substantially improved by conditioning diffusion models on structured semantic scene graphs. In sign language, discourse syntax operates as a dynamic 3D spatial scene graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where nodes represent spatial coordinate loci (L_1, L_{3a}, L_{3b}) and edges represent directional agreement predicates. Adapting her scene-graph conditioning methods via Relational Graph Transformers enables motion denoisers to be explicitly grounded in discourse-level spatial relations, resolving directional agreement verbs without spatial drift.
2. **Topologically and Geometrically Constrained Diffusion (Extending SCOPE / Graph Continuity):** In surgical video generation and segmentation (*HieraSurg* [7], *SCOPE* [6]), Dr. Farshad's work proves that biological structures require explicit graph-based continuity regularizers. Articulated human hands and arms exhibit identical topological vulnerabilities during neural diffusion. Adapting these graph constraints to the 52-joint SMPL-X skeletal tree guarantees bone-length invariance and anatomical joint limits, preventing joint popping and finger distortions.

4 Phased Research Plan: Core Scope vs. Collaborative Horizons

To maintain a rigorous, realistic doctoral scope, the 4-year research plan is organized around a **3-Paper Cumulative Doctoral Strategy**, distinguishing core deliverables from collaborative stretch horizons:

Phase / Timeline	Core Focus Area	Key Research Activities & Target Venues
Years 1–2(Parallel Tracks A & B)	Kinematic Synthesis & Streaming Perception	- Ethical approval from Aalto University Research Ethics Committee; establish Deaf Advisory Panel with <i>Kuurojen Liitto ry</i>. - <i>Track A (Kinematics)</i>: Procedural arm IK scaffold seeded with 3D-LEX v1.0 [11]; Continuous Residual Diffusion with localized tolerance ($r_{tol} \leq 5$ mm). - <i>Track B (Perception)</i>: HaMeR hand tracking, sternum normalization, and $\mathcal{O}(1)$ Causal Spatial Memory Bank (M_t) resolving directional and orientation-only verbs ($\vec{v}_{aim}$). - Paper 1 Deliverable (M20): <i>Anatomically Constrained Residual Diffusion for Articulated Sign Kinematics</i>. Target: CVPR / ICCV.
Year 3(Core Foundation)	Spatial-SignLLM Foundation Modeling	- Deploy sub-lexical tokenizer and 3D coordinate projector ($\mathbf{z}_{locus}$). - Pretrain Spatial-SignLLM on CSC’s EuroHPC LUMI supercomputer (AMD MI250X nodes) across How2Sign [3] and Motion-X++ [16]. - Paper 2 Deliverable (M32): <i>Spatial-SignLLM for Continuous Sign Language Translation</i>. Target: ACL / EMNLP.
Year 4(Collaborative Horizons)	Unified Foundation Model & Open-Domain Chat	- Couple Spatial-SignLLM with residual diffusion for bidirectional interaction across ASL and Finnish Sign Language (SVK). - Benchmark on an Open-Domain Everyday Conversational Study with certified Deaf/SVK consultants from <i>Kuurojen Liitto ry</i>. - Paper 3 Deliverable (M42): <i>Multimodal Spatial Foundation Modeling for Open-Domain Assistive Communication</i>. Target: NeurIPS / ICLR. - Doctoral dissertation compilation and defense at Aalto University.

Table 1: Phased Doctoral Work Plan organized into a 3-Paper Cumulative Dissertation Structure.

4.1 Computational Feasibility & Ethical Governance

EuroHPC LUMI Supercomputer: Large-scale pretraining on Motion-X++ (19.5M frames) [16] will utilize CSC’s LUMI supercomputer in Kajaani, Finland. Training will run on LUMI-G AMD Instinct MI250X nodes via native PyTorch AMD ROCm / HIP and high-speed parallel Lustre scratch storage.

Participatory Co-Design: In adherence to “Nothing About Us Without Us,” an ongoing **Deaf Advisory Panel** will be formed with the Finnish Association of the Deaf (*Kuurojen Liitto ry*) from Month 1, providing qualitative guidance on avatar facial blendshapes and linguistic naturalism.

References

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